

Context Software architecture for Machine Learning Systems (MLSystems) that improves operational reliability for dynamic upstream environments remains unresolved. While Machine Learning (ML) artifacts perform well in controlled settings, they struggle to generalize to real-world applications, limiting their use in critical areas [1].

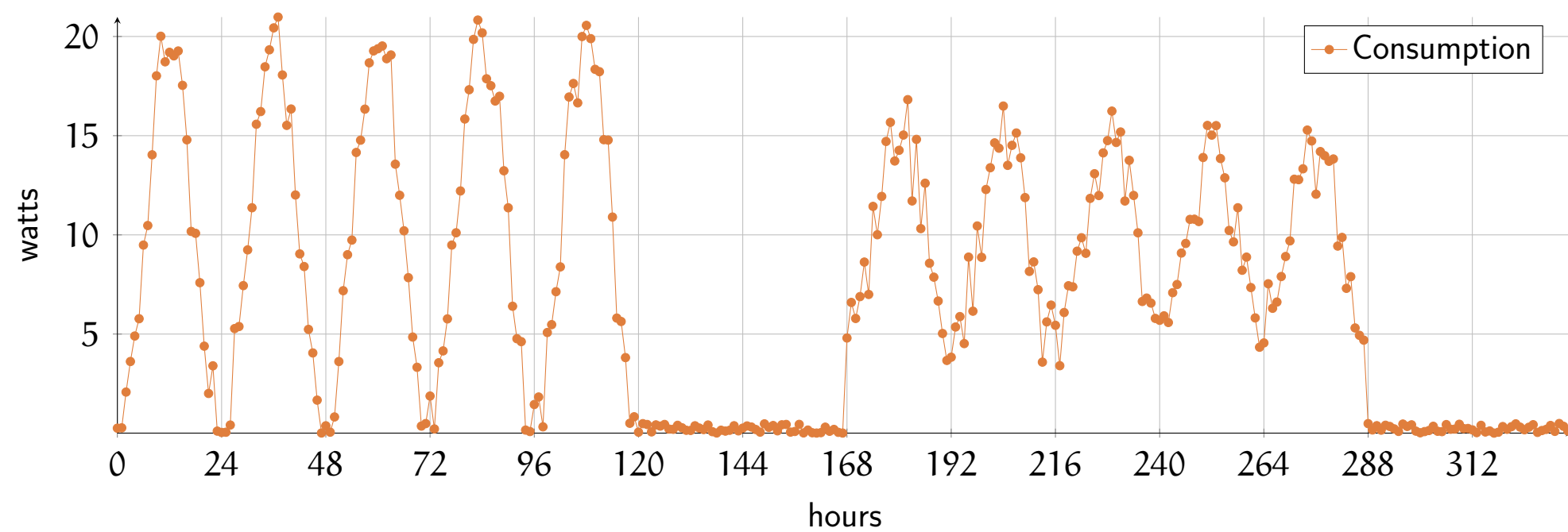


Figure 1: A synthetic upstream signal with dynamic variations over time [1, Fig. 2].

Methodology The survey uses a four-step (Fig. 2) process: 1) defining the search string, 2) extracting results from the database, 3) screening titles and abstracts with exclusion criteria, and 4) conducting a full-text eligibility assessment. This results in 17 papers that guide future research questions, as shown in Fig. 5.

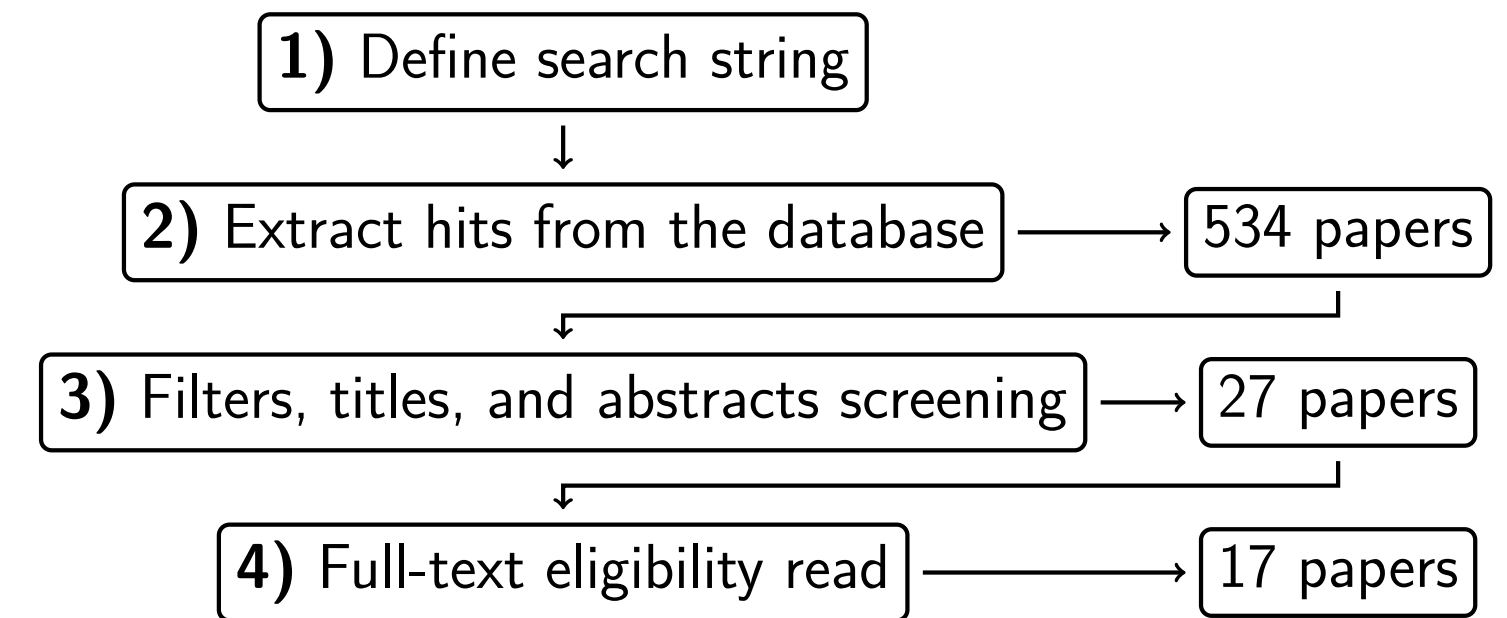


Figure 2: Illustration outlining the four-step selection process and the number of remaining papers at each stage. Figure adopted and modified from [1, Fig. 4].

What is the Transition from

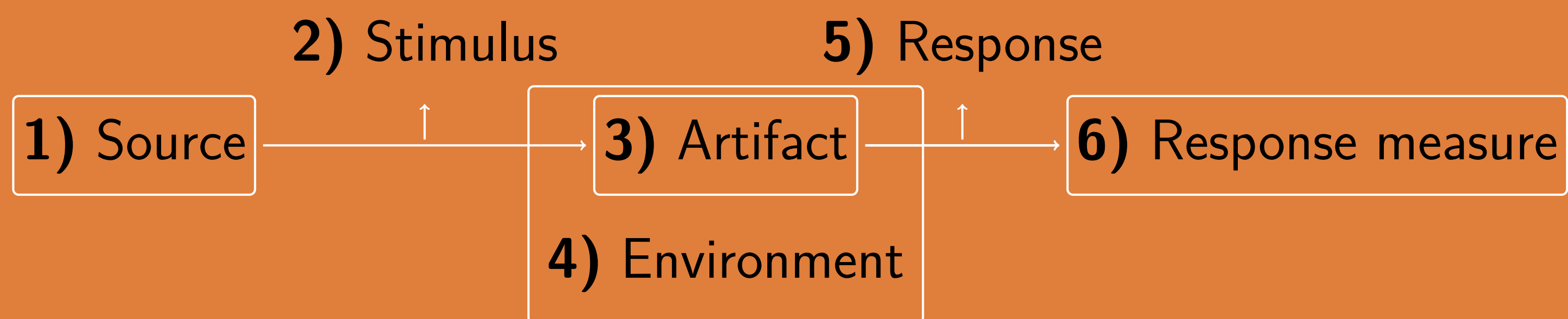


Figure 3: A general approach for outlining a quality attribute scenario applicable to any software system [2, Fig. 3.1].

To Quality Attributes for

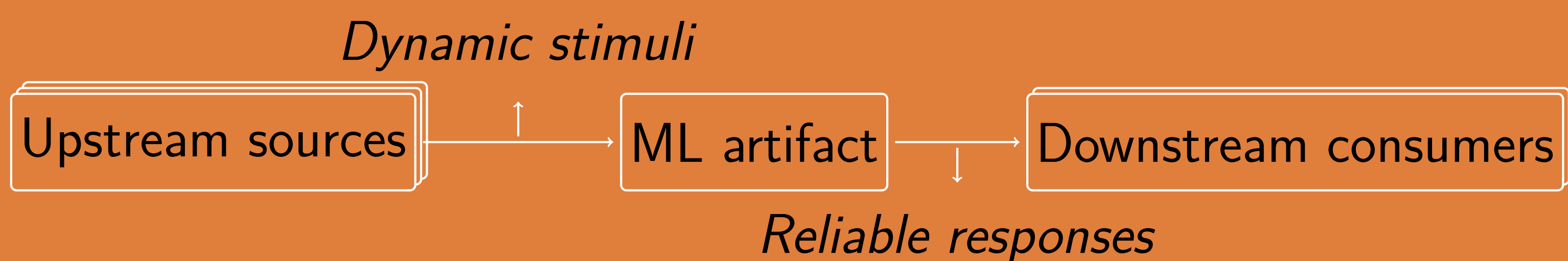


Figure 4: A modified illustration of a MLSystem composed of three components: Upstream sources, an ML artifact, and multiple downstream consumers [1, Fig. 1].

Machine Learning Systems?

Future Research Questions

- RQ1:** How do we define reliability as a quality of service metric for a MLSystem?
RQ2: What are the differences between the industries' training and operational metrics and environments?
RQ3: What are the important data streams to collect to compute operational metrics in a deployment?
RQ4: How do we automate the collection and computation of the operational metrics in a MLSystem pipeline?
RQ5: What is the engineering pipeline to optimize operational metrics for a MLSystem deployment?

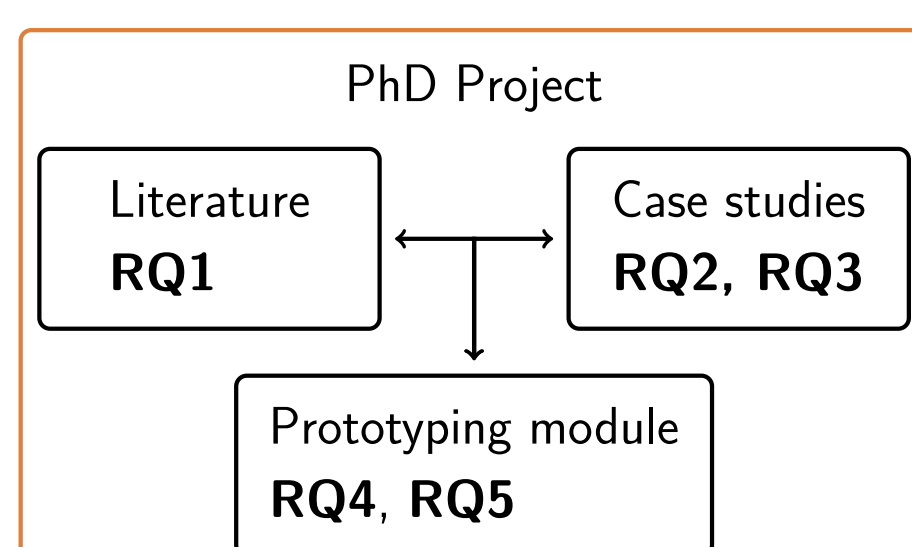


Figure 5: The connection between upcoming research inquiries and the direction of future studies. This figure is a modified version of [1, Fig. 8].

Knowledge Gap "The gap is the absence of adequate operational quality metrics for reliable MLSystems, which include the dynamics of upstream sources, ML artifacts, and downstream consumers in operation." - [1].

Supervisors

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References

- [1] A. Bæk-Petersen, "Survey on operational metrics for reliable machine learning systems," in *2025 IEEE 22nd International Conference on Software Architecture Companion (ICSA-C)*, Apr. 2025.
 [2] L. Bass, P. Clements, and R. Kazman, *Software Architecture in Practice*, 4th ed. Addison-Wesley Professional, 2021.